

PROFESSIONAL PORTFOLIO · 2026

MUHAMMAD FATHONI

Electrical Engineer · Master's Student

AI & IoT for Industry · Predictive Maintenance · Embedded Systems

Universitas Islam Indonesia · B.Eng. GPA 3.78 / 4.00 · M.Eng. in progress



fathonimuhammad2411@gmail.com

+62 822-5850-3481

Yogyakarta, Indonesia

[linkedin.com/in/muhammad-fathoni-038274266](https://www.linkedin.com/in/muhammad-fathoni-038274266)

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01 · ABOUT ME

About Me

I am an Electrical Engineer from Universitas Islam Indonesia (GPA 3.78, full scholarship), now continuing with a Master of Electrical Engineering at the same university. I work where hardware meets software: sensors, control panels and firmware on one side, machine learning, language models and web platforms on the other.

My focus today is AI and IoT for heavy industry, especially mining and energy. My latest project, SIAGA, reads vibration data from equipment, predicts failures and raises work orders and purchase requests on its own. On a real bearing failure dataset it warned about 25 hours before the machine stopped. I have also built multi agent AI systems, RAG chatbots with measurable guardrails, and shipped a mobile app to Android and iOS as a co-founder.

I learn fast, I like owning things end to end, and I care most about systems that people on site can actually trust.



CAREER OBJECTIVE

To join an industrial or technology team as an Engineer or AI specialist, applying my electrical engineering foundation, IoT experience and applied AI skills to make operations in mining, energy and manufacturing safer, more reliable and more efficient.

02 · EDUCATION

Education & Academic Records

Universitas Islam Indonesia (UII)

Sep 2026 – Present

Master of Electrical Engineering · Faculty of Industrial Technology

Universitas Islam Indonesia (UII)

Aug 2022 – May 2026

Bachelor of Electrical Engineering · Faculty of Industrial Technology

GPA 3.78 / 4.00

140 SKS Completed · UII Excellent Community Scholarship

KEY COURSEWORK

- Power Systems
 - Control Systems
 - PLC & SCADA
 - Renewable Energy
 - Power Electronics
 - Embedded Systems
- Machine Learning
 - Digital Systems (VHDL/FPGA)
 - Microprocessor Systems
 - Telecommunications
 - Computer Networks
 - Engineering Mathematics

03 · SKILLS

Technical Skills

HARD SKILLS

POWER & ELECTRICAL SYSTEMS

- Electrical Troubleshooting & Wiring
- Circuit Analysis (Analog & Digital)
- Power System Reliability (SAIDI/SAIFI)
- PLC Programming · PID Control · LabVIEW

EMBEDDED & IOT

- Microcontrollers (ESP32, Arduino, ATmega)
- Sensors (PZEM-004T, DHT, MQ Series)
- MQTT & Real Time Telemetry · Industrial Control Panels
- Proteus, CodeVision AVR, VHDL/FPGA

PROGRAMMING & DATA

- Python, C, C++, HTML, CSS, JS
- Machine Learning (scikit-learn, Isolation Forest, Random Forest)
- Generative AI: RAG, AI Agents, LLM Evaluation
- Claude, Gemini, Ollama · Pandas, Matplotlib

TOOLS & NETWORKING

- ERPNext, Docker, FastAPI, TimescaleDB, Firebase
- Computer Networking (HCIA-Datacom)
- Git · MATLAB · Microsoft Office

SOFT SKILLS

Critical & Analytical Thinking

Problem Solving

Teamwork & Collaboration

Communication & Public Speaking

Leadership & Coordination

Time Management

Adaptability & Fast Learning

Detail-Oriented & Initiative

LANGUAGES

Bahasa Indonesia — *Native*

English — *Professional Working Proficiency*

04 · PROJECTS

INDUSTRIAL AI · SOLE DEVELOPER · 2026

SIAGA: Predictive Maintenance ERP for Mining

Sole Developer (Architecture, ML, ERP Integration, IoT Pipeline)



- ERPNext
- Isolation Forest
- TimescaleDB
- MQTT
- Docker
- Ollama

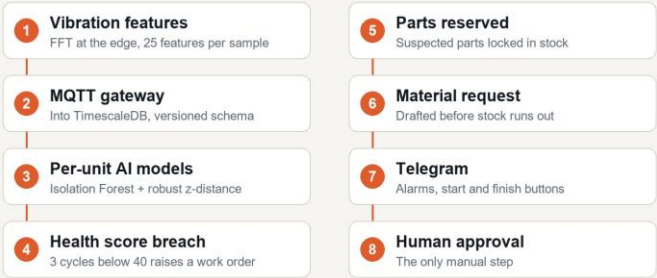
Condition based maintenance for mining equipment, built on ERPNext. Vibration data flows over MQTT into TimescaleDB, and an anomaly model scores every machine against its own healthy baseline. When the health score stays low, SIAGA raises a work order with the suspected parts, reserves the stock and drafts a purchase request. The only human step is approval.

KEY HIGHLIGHTS

- Warned about 25 hours before a real bearing failure (NASA IMS data), and 80 and 128 hours ahead on a second dataset
- Per unit Isolation Forest models, each scored against that unit's own healthy baseline
- Parts ordered before the alarm, based on the projected health trend
- Telegram bot for mechanics and a read only LLM assistant on local Ollama for planners

SIAGA

Condition-based maintenance that ends in a purchase request, on ERPNext



Detected failure 25 hours before the rig stopped (IMS bearing dataset) | docker compose up

How SIAGA turns vibration into a purchase request

04 • PROJECTS (CONT.)

SIAGA: From Sensor Data to Approved Purchase

Screens from the running system, replaying a real bearing failure dataset. github.com/Toni2411/siaga



Health score of a pump dropping over several days

E > SIAGA > SIAGA Work Order > WO-2026-00126
[Q Search or type a command \(Ctrl + G\)](#)
[Help](#)

Otomatis: kenaikan getaran lebar pada Pompa D... Selengkapnya

[Lihat:](#)
🔍
➤

Assigned To + Attachments + Tags + Share + FOLLOW Administrator last edited this 10 minutes ago SIAGA Bot created this 10 minutes ago	Jukka * Otomatis: kenaikan getaran lebar pada Pompa Dewatering Unit 01 Aset * ACC-ASS-2026-00001-1 Nama Aset Pompa Dewatering Unit 01 Katex Alat Pompa dewatering listrik Kategori COMP-00003 Pemicu ▼ Pekerjaan Urutan Pekerjaan Ditentukan otomatis oleh SIAGA. Sesi keesahian 38.71 pada 2026-09-26 15:19:58.650826. Part di bawah adalah dugaan dari gejala "forecast"; pastikan.
--	---

Work order raised automatically once the score crossed the threshold

Home > **Block** > **Material Request** > **000012020-000003** Search to view a historical CRM >... Help >...

Purchase Request for Bearing Remnant ZA-2115 Print Get Items From: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 101

Purchase request drafted before the alarm, with the reason written out

Mechanics start and close work orders from Telegram

04 · PROJECTS (CONT.)

FINAL PROJECT · 2025

SEMS: Smart Energy Management System

Sole Developer (Hardware, Firmware, ML, Web, AI Integration)



ESP32

IoT

Random Forest

Generative AI

Firebase

An end-to-end smart energy monitoring and recommendation platform that combines real-time power telemetry, predictive machine learning, and generative AI to help Indonesian households reduce electricity consumption. The system integrates ESP32-based hardware, a Random Forest forecasting model, Firebase real-time database, and Google Gemini 2.5 Flash to deliver context-aware energy advice.

KEY HIGHLIGHTS

- Real-time power monitoring with ESP32 + PZEM-004T + DHT22 sensors over MQTT
- Random Forest model achieved R^2 0.9893 (98.93% accuracy) for monthly forecasting
- Google Gemini 2.5 Flash integration for personalized energy-saving recommendations
- Firebase dashboard with PLN tariff calculator (450VA – 5500VA customer profiles)

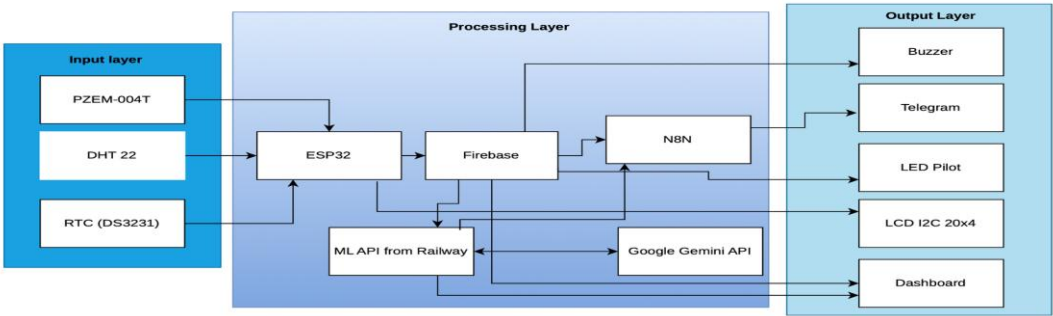


SEMS device — front panel with LCD display

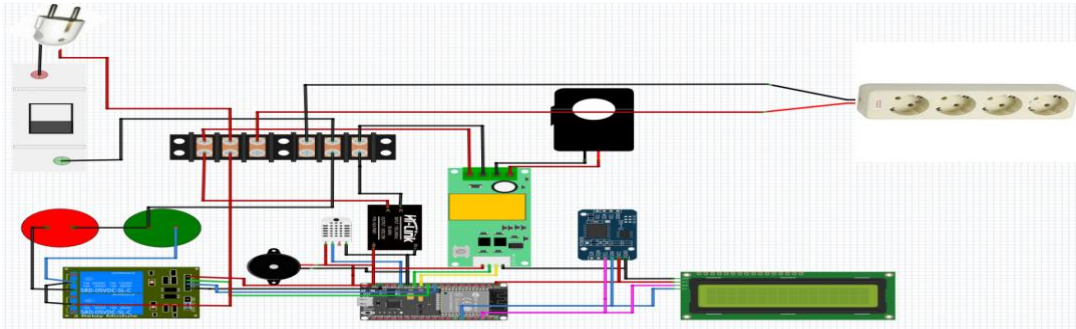
04 · PROJECTS (CONT.)

SEMS: System Architecture & Hardware

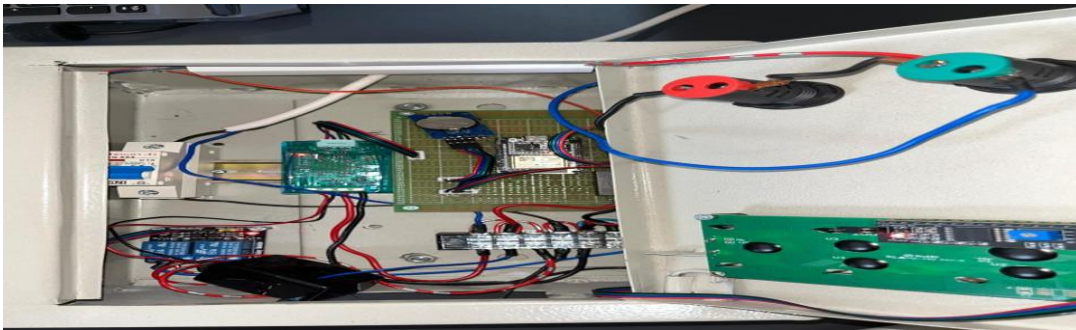
A complete IoT pipeline: from physical sensors and microcontroller to cloud database, ML inference, and AI-powered dashboard.



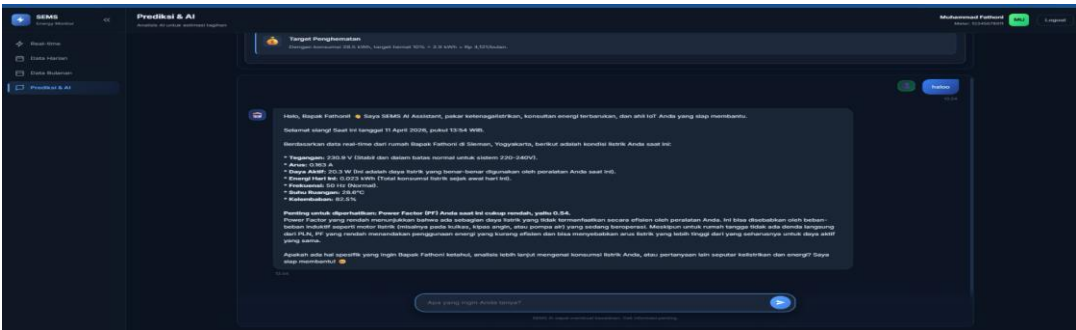
System architecture: Input → Processing → Output Layer



Hardware schematic — ESP32, PZEM-004T, MCB, relay, LCD



Internal wiring of the SEMS prototype enclosure



Real-time dashboard with SEMS AI Assistant (Gemini)

04 · PROJECTS (CONT.)

MULTI AGENT SYSTEM · SOLE DEVELOPER · 2026

Autonomous AI Video Studio

Sole Developer (Architecture, Agents, Prompts, Video Pipeline)

Claude API

Python

AI Agents

YouTube API

ffmpeg

Telegram

A team of AI agents that writes, films, checks, uploads and learns from an animated YouTube series, controlled from Telegram. I wanted a channel that runs itself without turning into low quality AI spam, so every agent owns one job and the others check its work.

KEY HIGHLIGHTS

- Supervisor agent scores each script against a rubric and sends it back for revision
- 36 AI shots per episode with one consistent character; a QC agent rejects bad frames and dishonest titles
- Uploads privately through the YouTube Data API, then learns from day 1, 3 and 7 metrics
- Every step reported to Telegram, with a monthly API budget that falls back to free models



Episode thumbnail and the 36 generated shots

04 · PROJECTS (CONT.)

LLM RELIABILITY · SOLE DEVELOPER · 2026

Guardrailed LLM Content: RAG Chatbot and n8n Generator



Sole Developer (Prompts, Workflow, Retrieval Pipeline, Evaluation)

Python

RAG

Google Gemini

n8n

Embeddings

Gradio

Three AI systems built around one question: what stops the model from confidently saying something false? A marketing post whose prompt forbids any figure the model cannot source, a weekly social media generator built as an n8n workflow, and a RAG chatbot written from scratch that answers from an archive of speeches.

KEY HIGHLIGHTS

- 21 node n8n workflow with a regex gate; anything it catches goes to human review
- RAG built by hand: chunking, embeddings, retrieval, reranking and a cap per document
- A calibrated similarity floor makes the bot refuse when nothing relevant is retrieved
- Faithfulness 1.00 across 15 evaluation cases, with no invented quotations or dates

Guardrailed LLM Content

Three AI systems built around one rule: never state what you cannot source

1

Grounded prompt
Forbids any figure it cannot source

2

Post generator
21 node n8n workflow, weekly

3

Regex gate
Deterministic check after the prompt

4

Human review
Anything caught waits for a person

5

RAG from scratch
Chunking, embeddings, retrieval

6

Relevance floor
Refuses when nothing fits

7

Calibrated
Floor measured, not guessed

8

Evaluated
Faithfulness 1.00 on 15 cases

Python | Google Gemini | n8n | Embeddings | Gradio

RAG Chatbot Evaluation

15 test cases, LLM judge, calibrated relevance floor at 0.59

Faithfulness

1.00

No invented quotations or dates

Refusal accuracy

3 / 3

Out of scope questions declined

Answer relevancy

0.87

Mean across 12 graded answers

Stopped by the floor

2 / 3

Refused before generation

Python | Gemini embeddings | LLM reranking | eval harness in repo

Guardrails and the measured evaluation results

04 · PROJECTS (CONT.)

INDUSTRIAL CONTROL · TEAM PROJECT · 2026

Wastewater Treatment Control Panel

Control Panel Firmware and Monitoring Dashboard



ESP32

C++

State Machine

Firebase

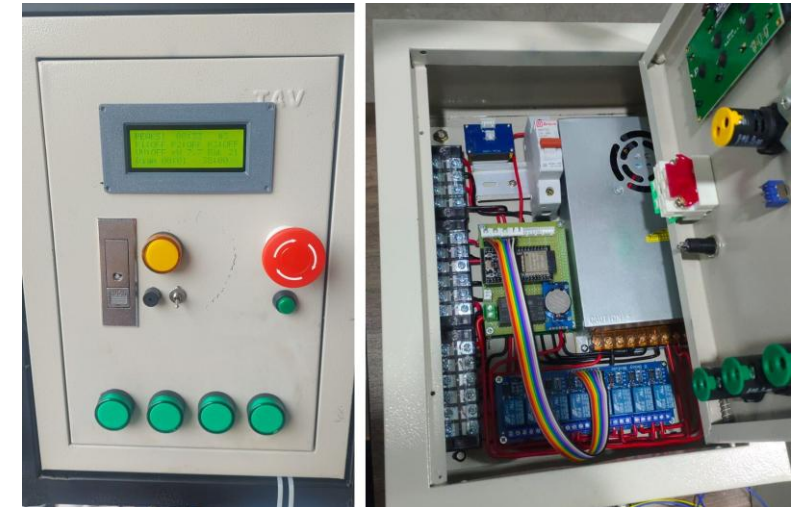
Chart.js

Wokwi

An ESP32 control panel that runs a small wastewater treatment unit on its own, in automatic 120 L batches, together with a web dashboard that lets operators watch it live. The panel sequences the pumps, reads water level and pH, catches faults and reminds the operator when the filter needs a backwash.

KEY HIGHLIGHTS

- State machine for fill, dose, react and treat stages, switching four pumps and a UV lamp
- Ultrasonic level and pH monitoring, fault handling, counters kept in non volatile memory
- Realtime dashboard with history charts, CSV export and an event log
- Logic validated in a Wokwi simulation against a written test procedure



The finished panel running a batch, and the wiring inside

04 · PROJECTS (CONT.)

COMMUNITY SERVICE · IOT · 2026

Mosha AirQ: Smoke Detection for Schools

Developer (Device, Firmware, Dashboard, Data Analysis)



ESP32

MQ-135

Firebase

JavaScript

Python

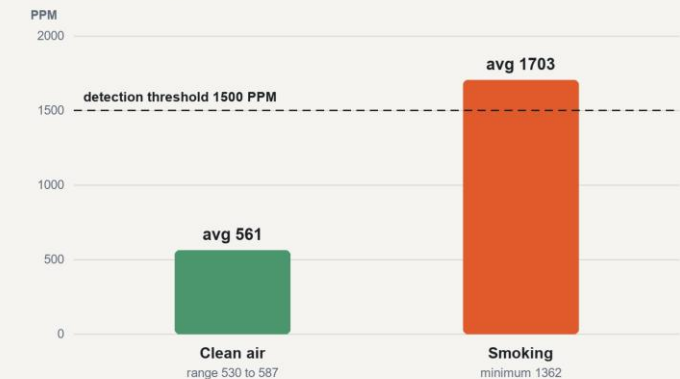
A low cost IoT air monitor that helps a senior high school enforce its smoke free policy. An ESP32 with an MQ-135 gas sensor classifies indoor air as safe, warning or smoking detected and reports it in real time, so teachers get an objective early warning instead of relying on patrols.

KEY HIGHLIGHTS

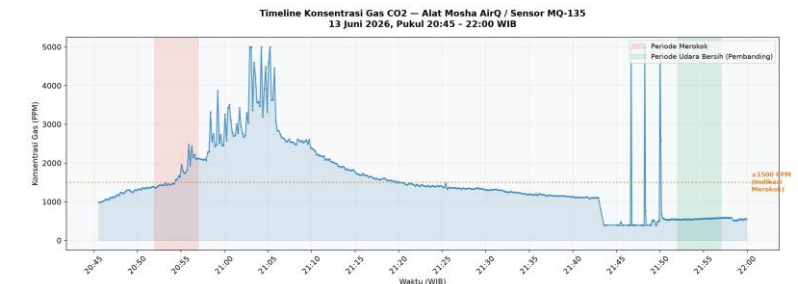
- 30 readings per condition: average gas level rose from 561 PPM in clean air to 1703 PPM
- The two ranges never overlapped, which set a detection threshold of 1500 PPM
- Rule based classifier on the device with a realtime web dashboard
- Written up as a journal article draft

Smoke Detection for Schools

MQ-135 gas readings, 30 samples per condition



ESP32 | MQ-135 | rule based classifier | realtime web dashboard



Clean air vs smoking, and the full test timeline

04 · PROJECTS (CONT.)

BACKEND ENGINEERING · 2026

Webhook Delivery Service

Sole Developer

Python

FastAPI

httpx

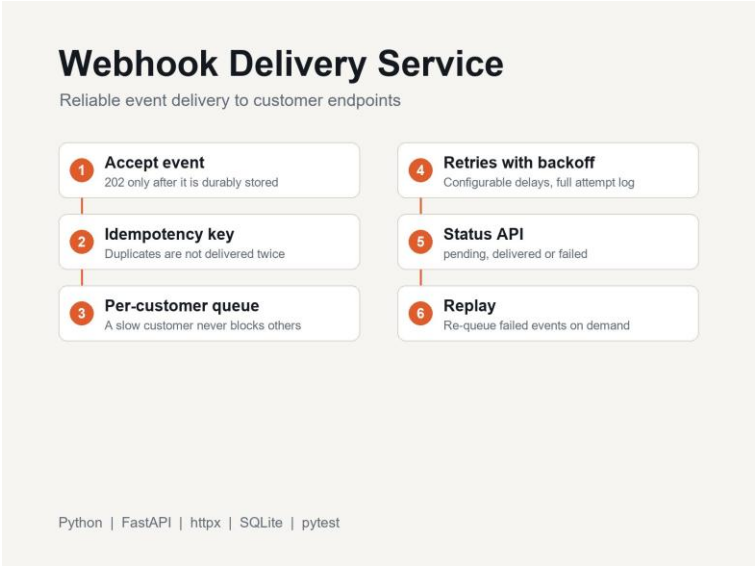
SQLite

pytest

A backend service that takes events from internal producers and reliably delivers them to HTTP endpoints registered by customers. It is built for the unhappy path: slow customers, endpoints that go down, and events that arrive twice.

KEY HIGHLIGHTS

- Events acknowledged only after they are durably stored; idempotency keys stop duplicates
- Each customer gets its own queue, so one slow endpoint never blocks the others
- Retries with configurable backoff, full attempt log, status API and replay
- Covered by pytest and a demo of healthy, flaky and down customers side by side



```
webhook delivery - demo run

$ uv run python demo.py
Sent 10 events to each of: healthy, flaky, down
t= 1s healthy: {'delivered': 10} flaky: {'pending': 5, 'delivered': 5} down: {'pending': 10}
...
t= 8s healthy: {'delivered': 10} flaky: {'failed': 2, 'delivered': 7, 'pending': 1} down: {'pending': 10}
t= 9s healthy: {'delivered': 10} flaky: {'failed': 2, 'delivered': 8} down: {'pending': 10}
...
t=20s healthy: {'delivered': 10} flaky: {'failed': 2, 'delivered': 8} down: {'pending': 10}

flaky endpoint received 8 unique events (0 duplicates)
down endpoint, first event: status=pending attempts=2 last_error=ConnectError (refused)

$ uv run pytest -q
13 passed in 2.17s
```

Delivery guarantees and a live demo run

04 · PROJECTS (CONT.)

PERSONAL PRODUCT · NOORA DEEN TECHNOLOGY · 2026

Noora: Muslim Habit-Building App

Co-Founder & Sole Engineer (Mobile, Backend, UI)

Capacitor 8

JavaScript

Supabase

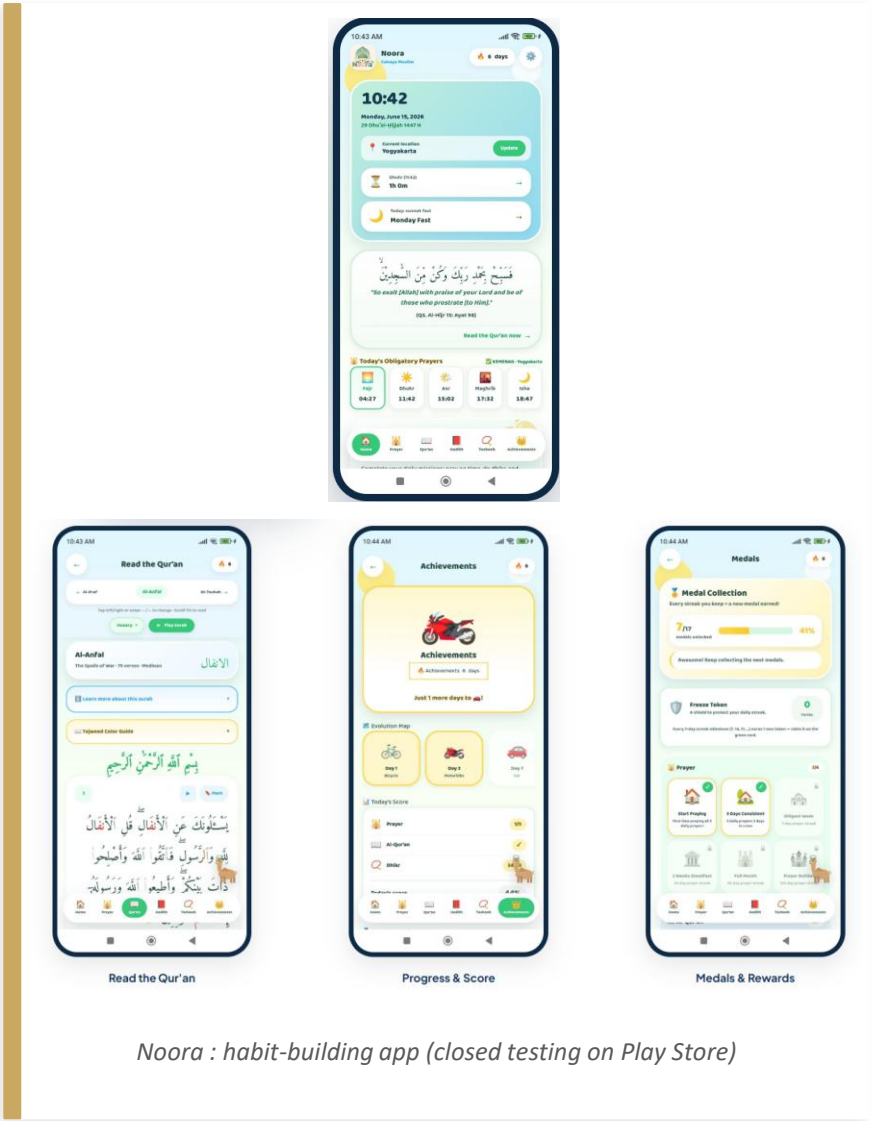
REST API

Android

A mobile application that helps Muslims build consistent daily worship habits in a positive, engaging way. Noora combines an international prayer-time engine, a complete digital Qur'an with an authentic 604-page Madinah mushaf mode, and a gamified “Garden” that grows with the user's consistency. Built as the flagship product of Noora Deen Technology, it runs fully online and offline, and is currently in closed testing on the Google Play Store.

KEY HIGHLIGHTS

- Region-aware prayer times: Kemenag for Indonesia, auto-switching to JAKIM, MUIS, Umm al-Qura, ISNA or Diyanet abroad
- Full Qur'an with verified Saheeh International translation, tafsir, and a 604-page Madinah mushaf with page-turn animation
- Gamified habit engine: per-category and combined streaks, freeze tokens, an evolving garden and claimable achievements
- Works fully offline, or syncs progress across devices in real time via Supabase; custom real-time ID/EN i18n engine



Noora : habit-building app (closed testing on Play Store)

04 · PROJECTS (CONT.)

CORPORATE WEBSITE · NOORA DEEN TECHNOLOGY · 2026

Noora Deen Technology: Company Website

Designer & Developer (Front-End, Branding)

HTML

CSS

JavaScript

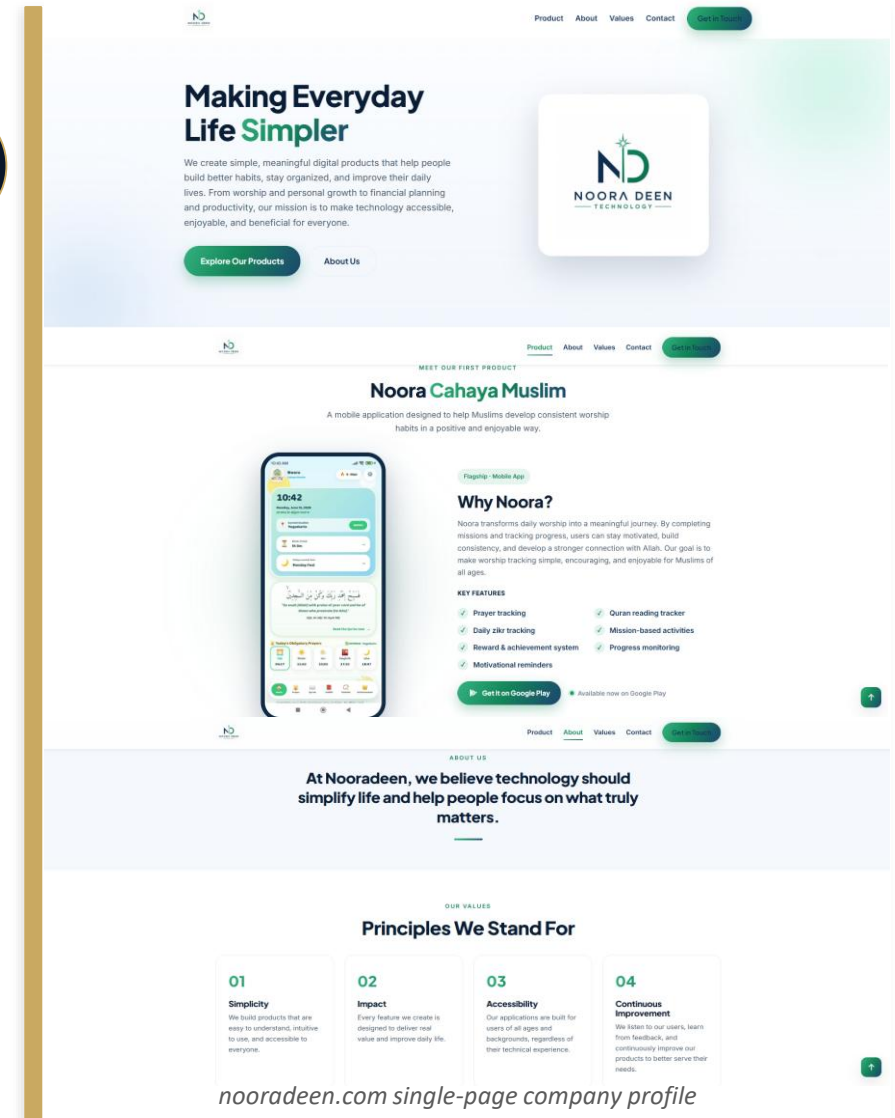
Responsive

Accessibility

A lightweight, fast single-page company profile for Noora Deen Technology, built entirely with vanilla HTML, CSS and JavaScript, no framework, no build step. The site presents the company's mission ("Making Everyday Life Simpler"), its products, values and contact channels, and is engineered for accessibility and smooth performance on any device.

KEY HIGHLIGHTS

- Fully responsive with a sticky navbar, hamburger menu and scroll-spy navigation across desktop, tablet and mobile
- Scroll-reveal animations and animated statistic counters built without any external animation library
- Accessibility-first: keyboard navigation, ARIA labels, and respect for prefers-reduced-motion
- Custom brand system: navy-and-green palette, north-star logo with an "ND" monogram, zero-backend contact form



nooradeen.com single-page company profile

04 · PROJECTS (CONT.)

PERSONAL PRODUCT · AI FINANCE · 2026

Saku: AI-Powered Personal Finance Manager

Sole Developer (Front-End, Backend, AI, Mobile)

React 19

TypeScript

Gemini 2.5

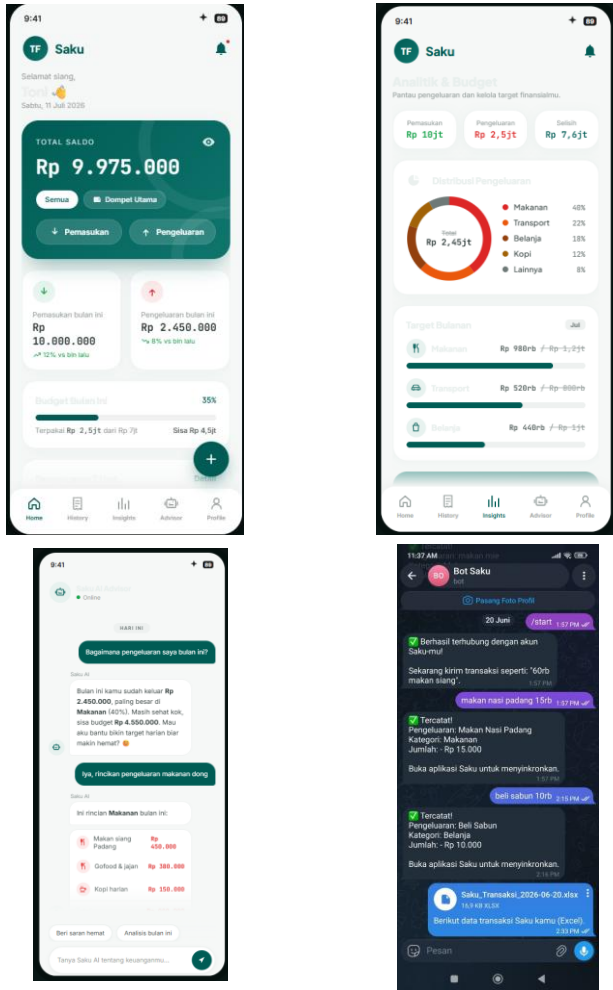
Express

Capacitor

A mobile-first personal finance manager for Indonesian users. Saku makes logging income and expenses nearly frictionless, then layers on an AI advisor (Google Gemini 2.5 Flash) that is grounded on the user's own transactions and replies in natural Bahasa Indonesia. A Telegram bot lets users record spending simply by chatting, “60rb makan siang” becomes a categorised expense of Rp 60.000.

KEY HIGHLIGHTS

- Telegram bot logs spending from natural language: “60rb makan siang” becomes a categorised Rp 60.000 expense
- Grounded AI advisor: WhatsApp-style chat answering questions about the user's real transactions in Bahasa Indonesia
- Analytics suite: category donut, 7-day income-vs-expense trends, budget progress and AI-generated insights
- Cross-platform React 19 + Vite client wrapped with Capacitor 8, deployed serverless on Vercel with Excel export



Saku dashboard, analytics and AI advisor

04 · PROJECTS (CONT.)

MACHINE LEARNING COURSE · 2024

Iris Flower Classification with Support Vector Machine

Developer



Python

SVM

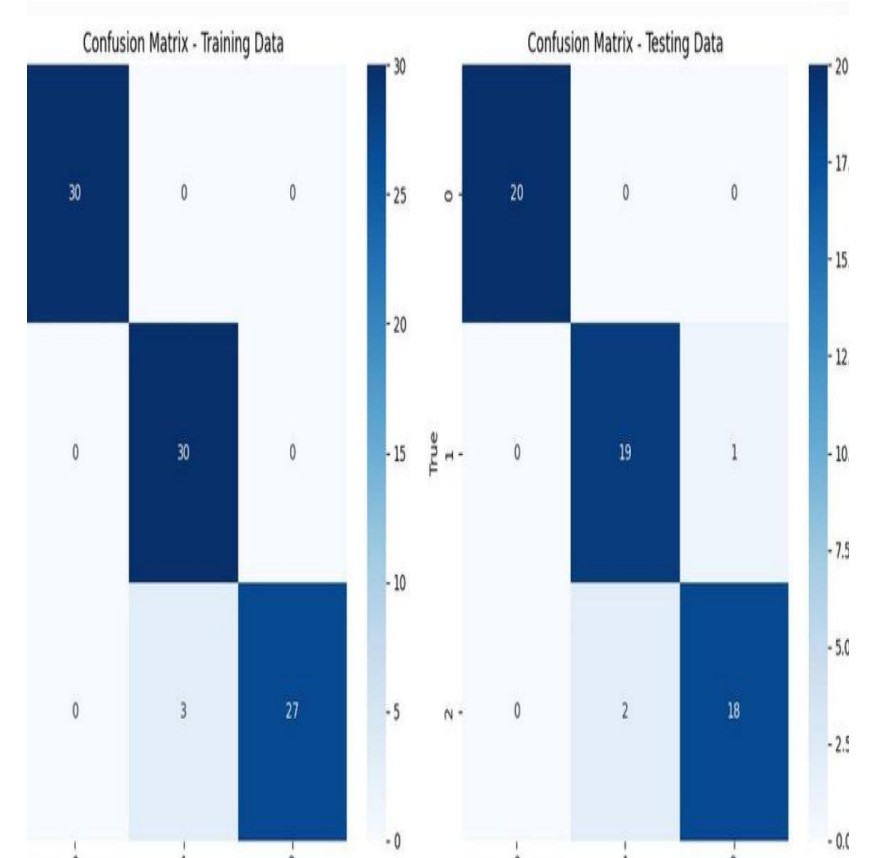
Scikit-learn

Google Colab

Built and evaluated a Support Vector Machine model to classify Iris flower species using petal and sepal measurements. The project followed a complete ML pipeline — from exploratory data analysis and preprocessing through model training, evaluation with confusion matrices, and classification reporting — strengthening my foundation in supervised learning workflows.

KEY HIGHLIGHTS

- Achieved over 90% classification accuracy on test set with linear-kernel SVM
- Train-test split of 80:20 with proper cross-validation methodology
- Full ML pipeline: preprocessing, training, evaluation, and reporting
- Practical foundation for advanced supervised learning workflows



Confusion matrix — training vs testing performance

04 · PROJECTS (CONT.)

PKM-KC FUNDED PROJECT · 2024

AromatheraWrist: Smart Bracelet for Anxiety Relief

Lead Developer · PKM-KC Team

Wearable

Biomedical

Mobile App

Embedded

An innovative wearable bracelet that integrates aromatherapy delivery with a real-time pulse sensor, controlled via a companion mobile application to help reduce anxiety symptoms in users. The project combined biomedical signal processing with user-centered design, and was awarded PKM Karsa Cipta funding by the Indonesian Ministry of Education.

KEY HIGHLIGHTS

- Awarded PKM Karsa Cipta funding by Indonesian Ministry of Education (2024)
- 4th Place — PKM IC Karsa Cipta Scheme
- Real-time pulse rate monitoring with embedded sensor system
- Mobile app integration for aromatherapy spray control



AromatheraWrist prototype — aromatherapy spray and pulse sensor

04 · PROJECTS (CONT.)

ELECTRONICS SYSTEMS PROJECT · 2024

IoT Automatic Temperature Control

Developer

ESP32

MQTT

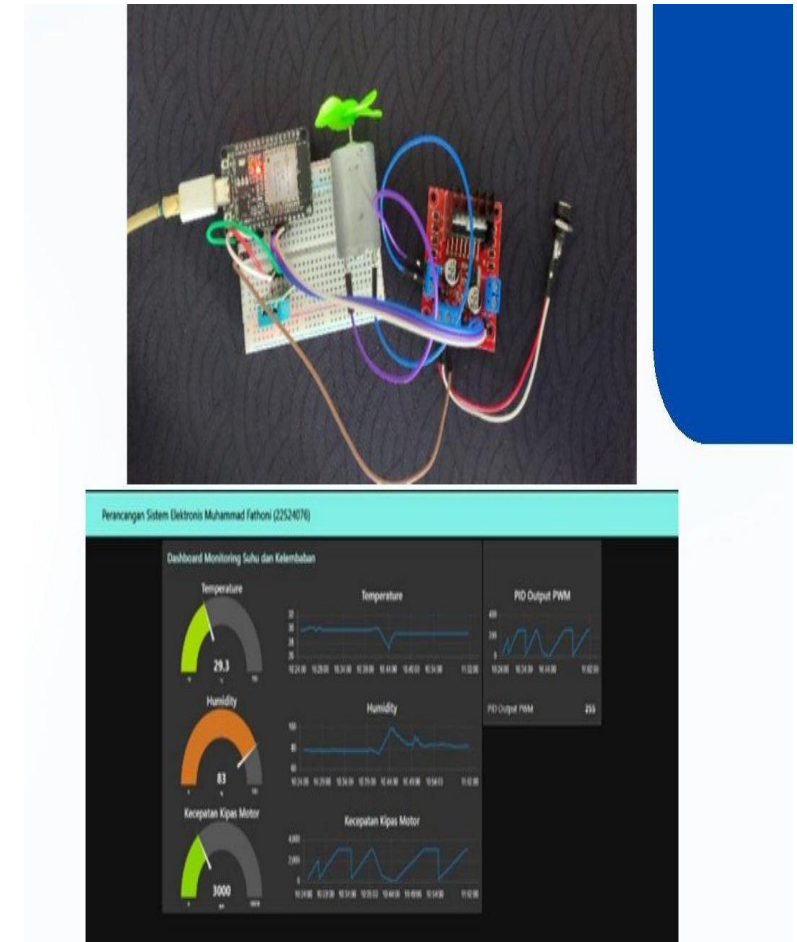
PID Control

Python

An IoT-based automatic temperature control system using ESP32 and DHT11 sensor, with a Python-based PID controller communicating via MQTT to dynamically regulate fan speed based on real-time room temperature. The system demonstrated stable closed-loop control with accurate response to environmental changes.

KEY HIGHLIGHTS

- ESP32 (publisher) + Python application (subscriber) over MQTT broker
- L298N motor driver controlling DC fan with PID-tuned response
- Real-time monitoring dashboard with temperature, humidity, and fan speed gauges
- Demonstrated stability and accuracy in real-time control loop



Hardware setup + monitoring dashboard with PID output

04 · PROJECTS (CONT.)

EMBEDDED SYSTEM FINAL PROJECT · 2024

Air Quality Monitoring Tool

Developer

Arduino Uno

NodeMCU ESP8266

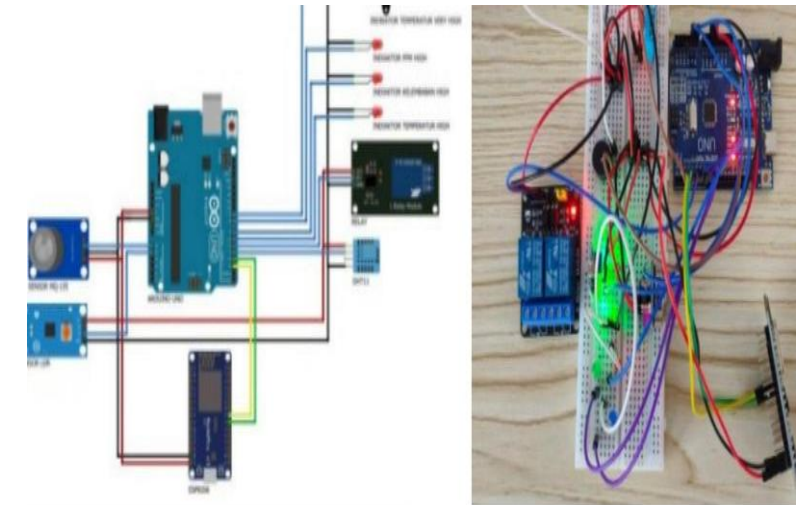
Blynk IoT

Multi-Sensor

A real-time air quality monitoring system that measures temperature, humidity, gas concentration (PPM), and light intensity using multiple sensors integrated with Arduino Uno and NodeMCU ESP8266. Data is transmitted to the Blynk IoT platform, enabling real-time monitoring and remote control via smartphone application.

KEY HIGHLIGHTS

- Multi-sensor integration: DHT11 (temp/humidity), MQ-135 (gas), LDR (light)
- Cloud-based dashboard with mobile remote control via Blynk IoT
- Real-time environmental data logging with notification alerts
- Day/night mode detection with automatic relay switching



Hardware schematic, breadboard prototype, and Blynk app

04 · PROJECTS (CONT.)

MICROPROCESSOR SYSTEMS PROJECT · 2024

Three-Lane Adaptive Traffic Light Simulator

Developer

ATmega328P

CodeVision AVR

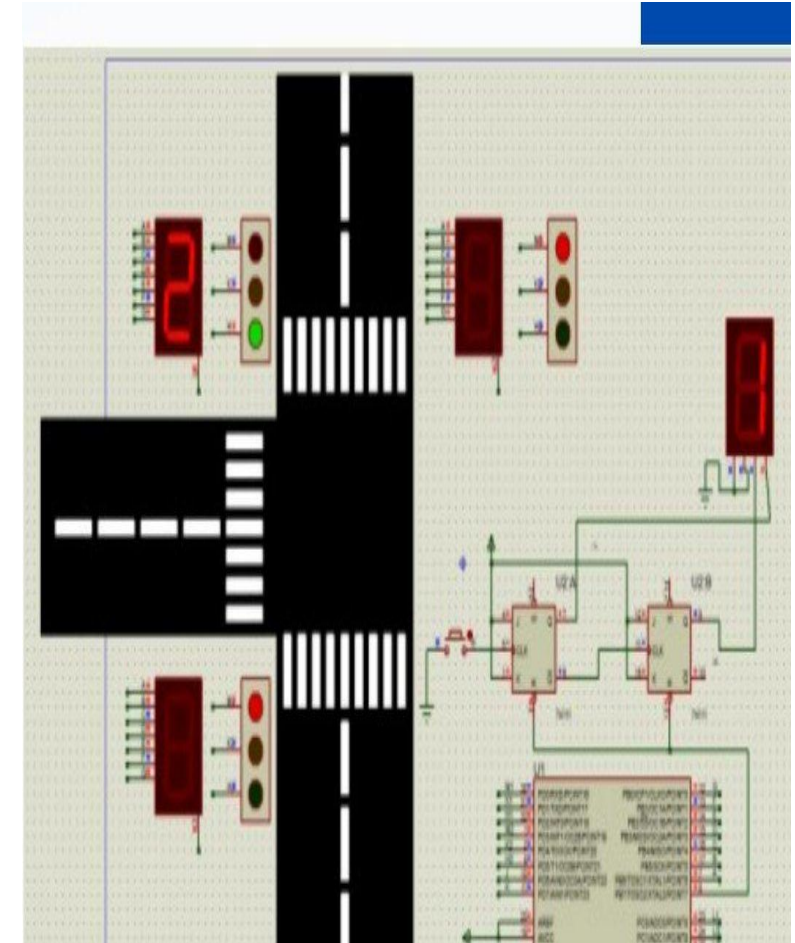
Proteus

C

An adaptive traffic light controller for a three-way intersection, programmed in C via CodeVision AVR (CVAVR) and simulated using Proteus 8 Professional. Each lane is equipped with a seven-segment countdown timer, and the system dynamically adjusts the green light duration based on traffic density signals from push buttons combined with JK flip-flop logic.

KEY HIGHLIGHTS

- ATmega328P microcontroller programmed in C via CVAVR
- 7-segment countdown display per lane with dynamic timing
- Density-aware green-phase scheduling using JK flip-flop logic
- Simulated and validated in Proteus 8 Professional



Three-way intersection simulation in Proteus 8

05 · EXPERIENCE

Professional Experience

- Full Stack / AI Engineer · Co-founder

Noora Deen Technology

- Built Noora, a cross platform Islamic lifestyle app with prayer times, the full Qur'an with per ayah audio and a streak based habit system.
 - Own the whole stack: offline first frontend, Supabase backend with realtime sync and row level security, and the Android and iOS release pipeline.
 - Designed and built the company website, nooradeen.com.

May 2026 – Present

Part time


- Engineering Intern

PT PLN (Persero) — ULP Pasir Pengaraian

- Worked in the Engineering (Teknik) Division, handling customer outage data through the APKT system.
 - Performed reliability analysis using SAIDI and SAIFI indices to evaluate distribution network performance.
 - Supported division performance evaluation and collaborated with field technicians on outage resolution.
 - Gained hands-on understanding of PLN distribution operations, work culture, and customer service standards.

Sep 2025 – Oct 2025

Riau, Indonesia



05 · EXPERIENCE (CONT.)

Teaching Experience

Laboratory Assistant — Telecommunication Systems

Department of Electrical Engineering, Universitas Islam Indonesia

- Guide undergraduate students through hands-on telecommunication systems experiments.
- Prepare lab modules, evaluate student reports, and assist with troubleshooting.

Apr 2025 – Jan 2026

Yogyakarta



Laboratory Assistant — Control Systems (PID & LabVIEW)

Department of Electrical Engineering, Universitas Islam Indonesia

- Mentored students on PID controller design, tuning, and LabVIEW-based simulation.
- Assisted in conducting practical sessions for closed-loop control experiments.

2025

Yogyakarta



05 · EXPERIENCE (CONT.)

Teaching Experience

Laboratory Assistant — Digital Systems (VHDL & FPGA)

Department of Electrical Engineering, Universitas Islam Indonesia

- Mentored students on digital logic design using VHDL and FPGA implementation.
- Supervised hands-on laboratory work in combinational and sequential logic circuits.

Laboratory Assistant — Electric Circuit II

Department of Electrical Engineering, Universitas Islam Indonesia

- Mentored second-year students on AC circuit analysis, network theorems, and resonance.
- Conducted lab assessments and supported instructors during practical sessions.



Lab session — undergraduate practical

TEACHING PROFICIENCY

Across four semesters as a laboratory assistant, I have mentored over 200 undergraduate students in core electrical engineering subjects — from fundamental circuit theory to modern control systems and digital design. This experience has sharpened my technical communication skills, patience, and ability to break down complex concepts.

06 · LEADERSHIP

Leadership & Organizations

IEEE STUDENT BRANCH UII

- **Director of Academies**
Jun 2024 – May 2026
- **Academies Staff**
Mar 2023 – Jun 2024
- **Member (IEEE Global)**
Jan 2024 – Dec 2024

UASC ELECTRIC VEHICLE

- **Head of Programming Division**
May 2024 – Mar 2025
- **Electrical Engineer Staff**
Dec 2023 – May 2024
- **Electrical Engineer Intern**
Jul 2023 – Dec 2023

UNISI ROBOTIC UII

- **Electrical Division Staff**
May 2024 – May 2026

UII EXCELLENT COMMUNITY (UII EC)

- **Full Scholarship Awardee**
Aug 2022 – May 2026

COMMITTEE EXPERIENCE

Electrical Link Togetherness (ELITE) — Master of Ceremony
Events Department · Department of Electrical Engineering UII
Hosted department-wide events; coordinated event flow and on-stage communication.

07 · ACHIEVEMENTS AND CERTIFICATIONS

Achievements

- 1ST

1st Place — IT Centrum Startup Hackathon 2024
Universitas Islam Indonesia · Dec 13–14, 2024
- 1ST

1st Place — Business Idea Competition
Student Entrepreneur Day 2024
- 4TH

4th Place — PKM IC Karsa Cipta Scheme
Ministry of Education · 2024
- ★

PKM Karsa Cipta Funding Recipient
Ministry of Education · 2024
- 3RD

3rd Place — PKM IC Community Service Scheme
Ministry of Education · 2025
- 5TH

5th Place — PKM IC Innovative Work Scheme
Ministry of Education · 2025
- ★

UII Excellent Community Full Scholarship Awardee
UII · 2022 – 2026

Certifications

AI Fundamentals: Foundations for Understanding AI IBM SkillsBuild · credly.com/go/NFH2IvaK	Sep 2026
Introduction to Large Language Models Google Cloud Skills	Sep 2026
Introduction to Generative AI Google Cloud Skills	Sep 2026
MATLAB Onramp MathWorks	Sep 2026
HCIA-Datacom V1.0 Huawei Talent Online	Dec 13, 2023
Basic PLC Training Dept. of Electrical Eng., UII	Dec 21, 2024
Python for Data Science (Beginner) Dept. of Electrical Eng., UII	Nov 2024 & May 2025
Leadership Training UII EC 2022 UII Student Affairs Directorate	Jun 29, 2024
Personal Development Basic Training UII Student Affairs Directorate	Jun 25, 2023
IEEE Leadership Class 2022 IEEE Student Branch UII	Nov–Dec 2022

08 · CONTACT

Let's Work Together.

I'm open to engineering, AI/ML and industrial AI roles, especially in mining and energy.

EMAIL

fathonimuhammad2411@gmail.com

PHONE

+62 822-5850-3481

LINKEDIN

linkedin.com/in/muhammad-fathoni-038274266

LOCATION

Yogyakarta, Indonesia · Open to remote and relocation

THANK YOU FOR YOUR CONSIDERATION
